

Car Claim Insurance

Business Report

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1. Introduction

The car insurance industry operates in an increasingly competitive and data-driven environment where the accurate assessment of risk, the optimisation of premium structures, and the effective management of claims are critical determinants of profitability and long-term sustainability. Insurance companies that leverage data analytics to understand their customer base and claims behaviour are better positioned to make informed underwriting decisions, design targeted policy products, and maintain adequate financial reserves.

This report presents a comprehensive analysis of a car insurance claims dataset comprising 7,647 policyholder records, conducted using Tableau as the primary analytical and visualisation platform. The analysis explores key relationships between customer demographic characteristics, vehicle attributes, and claims behaviour, to generate actionable business insights that support strategic decision-making across pricing, underwriting, product development, and customer retention functions.

The report is structured to present findings progressively, beginning with an examination of the customer demographic profile, followed by an analysis of the income and education landscape, vehicle characteristics, claim severity patterns, and risk segmentation by occupation. Each section concludes with specific business recommendations designed to improve the company's operational efficiency, profitability, and market competitiveness.

2. Project Objective.

The main aim of the project is to analyse the dataset of car insurance using Tableau. The analysis and recommendations will cater towards the business objective of a Typical Insurance Company providing car insurance to both personal and commercial users.

2.1. Assumptions:

1. Several factors are provided in the dataset. However, there may be other unseen factors that are not considered.
2. YOJ and birthdate will lead to the number of years/age computed as at 2020.
3. The dashboard presented is intended to analyse the car insurance claims in detail, in order for the insurance company to have an informed understanding of their customer base and to provide better customer service.

3 Data Dictionary:

The dataset has data on 10284 clients' car insurance information; it includes the customer demographic information (birthdate, income, value of house owned, marital status, gender, education level, occupation, travel time), the insurance claim information (value of car, claim amount, claim frequency, age of car, etc)

Public

Link:https://public.tableau.com/app/profile/samiksha.singh6604/viz/Book22_17789133965530/InsuranceClaimsAnalysis?publish=yes

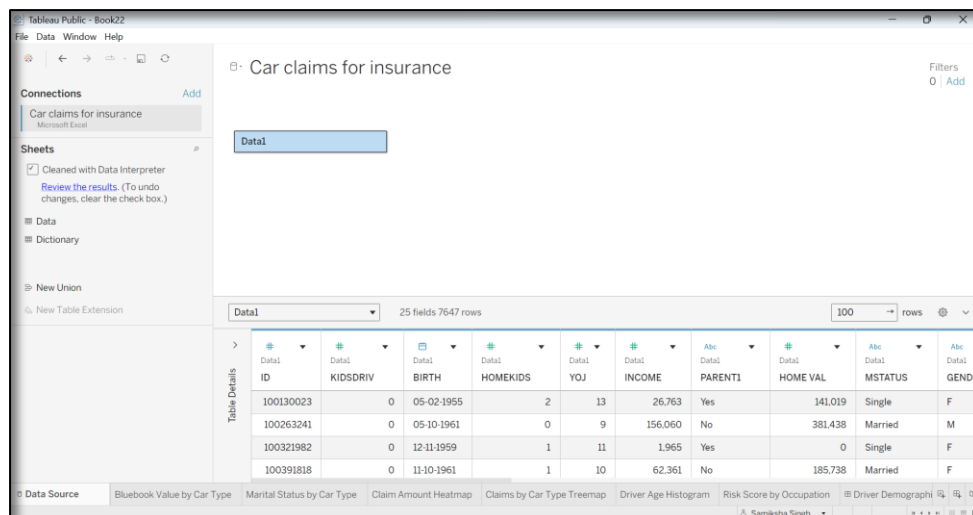


Figure 1: Data Source Connection — Car Claims for Insurance Dataset Connected via Microsoft Excel in Tableau Public

3.1. Data Source and Preparation

Dataset Overview: The analysis draws on the Car Insurance Claims dataset, encompassing 7,647 policyholder records across 25 fields. The dataset was connected to Tableau Public via Microsoft Excel and processed using Tableau's built-in Data Interpreter to ensure consistency and integrity throughout the analysis.

Dataset Fields Include: The dataset spans a broad range of variables — from core demographic attributes such as ID, birth date, gender, marital status, education, occupation, and income, to insurance-specific fields including home value, claim amount, claim frequency, car age, Bluebook value, vehicle use, and car type.

Data Cleaning: The Data Interpreter was applied at the point of source connection to detect and resolve formatting inconsistencies within the Excel file. To improve clarity across visualisations, Marital Status values were relabelled from Yes/No to Married/Single, making the data more intuitive and reader-friendly throughout the dashboards.

4. Creation of New Calculated Fields

To enhance the dataset and enable meaningful business segmentation, four new calculated fields were created directly within Tableau. Rather than relying solely on the raw data, these fields derive additional dimensions and measures that deepen analytical capability — forming the backbone of the dashboard visualisations and the business insights drawn throughout this report.

4.1. Age group computation

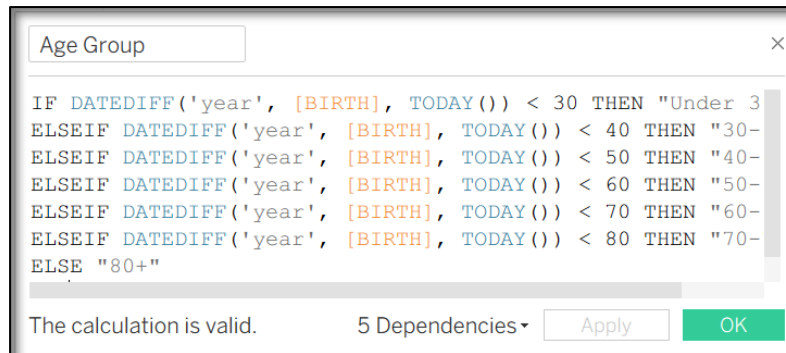


Figure 2: Age Group Calculated Field Formula in Tableau

- The Age Group field assigns each policyholder to a defined age band based on their recorded birth date.
- The DATEDIFF function calculates the difference in years between the policyholder's birth date and the current date.
- The resulting values are mapped to seven categories: Under 30, 30–39, 40–49, 50–59, 60–69, 70–79, and 80 and above.
- This field enables consistent age-based segmentation across all dashboards.
- It allows management to identify which age groups make up the largest share of the customer portfolio.
- Risk profiles can be assessed and compared across different age bands.
- The segmentation also supports the design of targeted policy products tailored to the needs of each demographic group.

4.2. Education Filter Calc

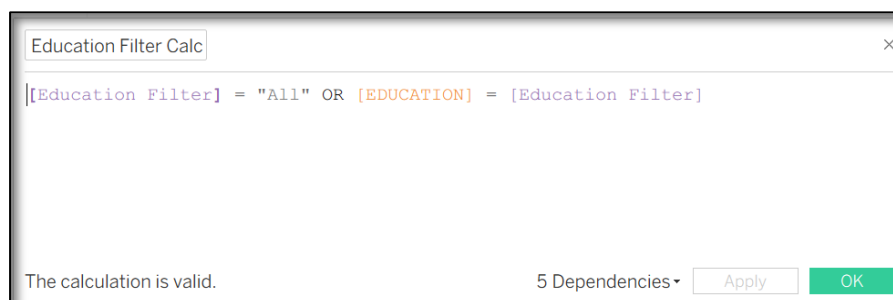


Figure 3: Education Filter Calc Calculated Field Formula in Tableau

- The Education Filter Calc is a boolean calculated field that works alongside the Education Filter parameter to enable dynamic filtering across relevant worksheets.
- The formula first checks whether the Education Filter parameter is set to "All," in which case records across all education levels are returned.
- If a specific education level is selected by the user, the formula filters the data to display only records matching that selection.
- This field introduces interactive filtering capability across dashboards without altering the underlying dataset.
- It allows management and analysts to isolate and examine the behaviour of specific education-level customer segments on demand.
- The dynamic nature of the filter makes it easy to switch between a broad portfolio view and a focused segment-level analysis.

4.3. Claim Amount Filter



Figure 4: Claim Amount Filter Calculated Field Formula in Tableau

- The Claim Amount Filter is a boolean calculated field that works in conjunction with the Min Claim Amount parameter to enable dynamic claim-based filtering across dashboards.
- The formula evaluates each record and checks whether the claim amount is greater than or equal to the value specified in the Min Claim Amount parameter.
- Records that fall below the defined threshold are excluded from the analysis, while those meeting or exceeding it are retained.
- This field allows analysts to shift focus exclusively toward significant claims, removing zero or low-value records that may obscure meaningful patterns.
- By filtering out lower-severity incidents, the dashboard delivers a clearer and more actionable view of high-severity claims.
- This supports better understanding of the incidents that carry the greatest financial impact on the business.

4.4. Risk Score



Figure 5: Risk Score Calculated Field Formula in Tableau

- The Risk Score is a composite measure that combines four key risk indicators: claim frequency, normalised claim amount, number of dependent drivers, and vehicle age.
- Each component is assigned a weight that reflects its relative contribution to overall policyholder risk.
- Together, these weighted inputs produce a single numerical score that allows direct and consistent comparison of risk levels across customer segments and occupational groups.
- The field underpins the Risk Analysis dashboard, serving as the primary measure for evaluating and ranking policyholder risk.
- It provides the foundation for occupation-based risk segmentation, making it possible to identify which customer profiles consistently generate elevated risk scores.

- The resulting insights enable the company to apply targeted premium adjustments and underwriting interventions where they are most needed.

5. Creation of New Parameters

Two parameters were created within Tableau to enable dynamic and interactive filtering across the dashboards. These parameters empower users to customise their view of the data in real time, enabling more targeted analysis and supporting data-driven decision-making at all levels of the organisation.

5.1. Education Filter

Figure 6 shows the configuration for the 'Education Filter' parameter in Tableau. The parameter is named 'Education Filter' and is of type 'String' with a display format of 'All'. The current value is set to 'All', and the value when the workbook opens is set to 'Current value'. The allowable values are configured as a list with six items: '<High School', 'Bachelors', 'High School', 'Masters', 'PhD', and 'All'. The parameter is set to 'Fixed' and 'When workbook opens'.

Value	Display As
<High School	<High School
Bachelors	Bachelors
High School	High School
Masters	Masters
PhD	PhD
All	All

Figure 6: Education Filter Parameter Configuration in Tableau

- The Education Filter is a string-type parameter configured as a list with six allowable values: All, Less than High School, High School, Bachelor's, Master's, and PhD.
- "All" is set as the default value, ensuring the full dataset is displayed unless a specific selection is made.
- The parameter operates in conjunction with the Education Filter Calc calculated field to dynamically filter all relevant dashboard visualisations.
- When a specific education level is selected, only records corresponding to that level are displayed across the affected worksheets.
- This parameter gives management the ability to isolate and examine the behavioural and financial characteristics of individual education-level segments.
- The insights generated support more informed and targeted premium pricing decisions.
- It also aids product development strategies by highlighting how customer needs and risk profiles vary across different education groups.

5.2. Min Claim Amount

The screenshot shows the 'Edit Parameter [Min Claim Amount]' dialog box. The 'Name' field contains 'Min Claim Amount'. Under 'Properties', the 'Data type' is 'Integer' and the 'Display format' is '0'. The 'Current value' is '0'. The 'Value when workbook opens' is set to 'Current value'. Under 'Allowable values', the 'Range' option is selected. The 'Range of values' section shows 'Minimum' as '0', 'Maximum' as '85,000', and 'Step size' as '1,000'. The 'Fixed' option is selected for the range. There is an 'Add values from' button. At the bottom are 'Cancel' and 'OK' buttons.

Figure 7: Min Claim Amount Parameter Configuration in Tableau

- The Min Claim Amount is an integer-type parameter configured as a range spanning from \$0 to \$85,000, with a step size of \$1,000.
- It operates in conjunction with the Claim Amount Filter calculated field to dynamically exclude claim records that fall below the user-defined threshold.
- This allows analysts to shift focus exclusively toward high-severity claims that carry a material impact on loss ratios and reserve requirements.
- The parameter updates dashboard visualisations in real time as the threshold is adjusted, ensuring the analysis always reflects the selected claim range.
- It is particularly valuable for actuarial and senior management teams who need to examine patterns within the high-severity claim segment.
- Low-value and zero-value claims, which can mask meaningful trends, are effectively removed from view when the threshold is applied.
- This sharpens the analytical focus and supports more accurate assessment of the claims that pose the greatest financial risk to the business.

6. Dashboard Explained

6.1.1. Distribution of Drivers by Age Group

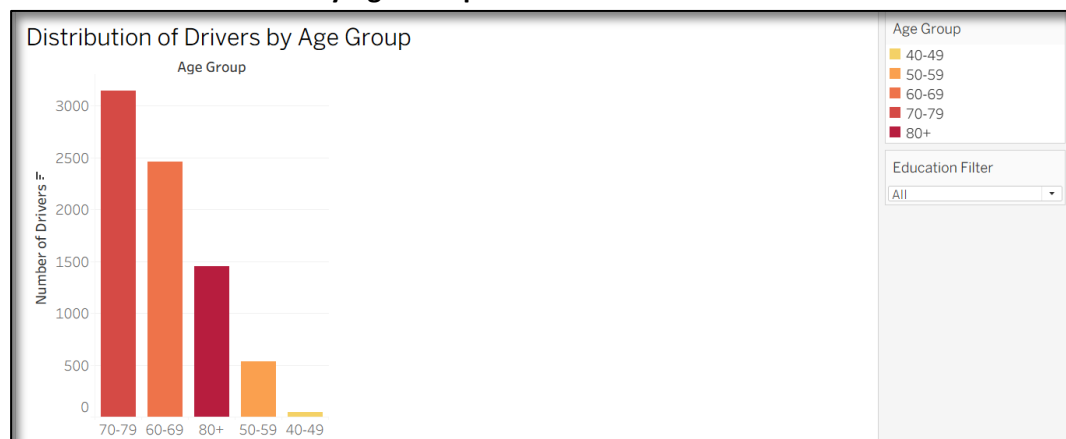


Figure 8: Distribution of Drivers by Age Group — Bar Chart

- The policyholder base is heavily concentrated among older drivers, with the 70–79 age group representing the single largest segment in the portfolio.
- Drivers aged 60 and above collectively account for more than 75% of the total customer base, indicating a strongly senior-skewed portfolio.
- Drivers under 50 are critically underrepresented, indicating a significant gap in acquiring younger customers.
- This age concentration elevates long-term claims exposure as the dominant cohort continues to age and associated health and mobility risks increase.
- To address this, the company should develop senior-specific policy products that incorporate relevant benefits such as medical assistance and roadside support.
- Simultaneously, targeted acquisition campaigns for younger drivers would help diversify the portfolio and reduce overreliance on a single age segment.
- Broadening the age distribution is essential to ensuring long-term revenue sustainability and a more balanced risk profile across the customer base.

6.1.2. Average Income by Education Level

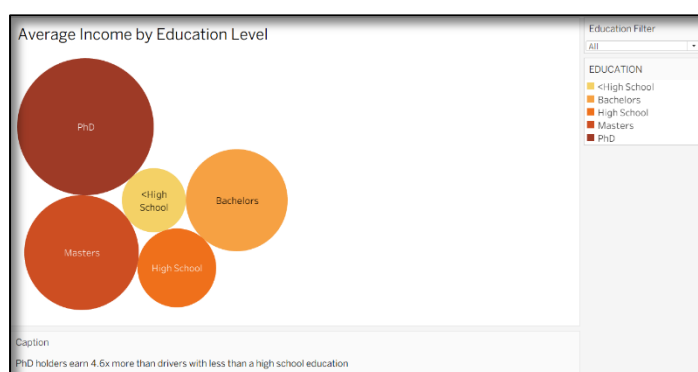


Figure 9: Average Income by Education Level

- A clear and consistent positive correlation exists between education level and average income across the policyholder base.

- PhD holders earn approximately 4.6 times more than those with less than a high school education, with average incomes of \$118,000 compared to \$26,000 respectively.
- This income gap is not isolated — it follows a steady upward trajectory across every education level, from the lowest qualification to the highest.
- Higher-income policyholders tend to own more expensive vehicles, which directly elevates potential claim values and increases reserve requirements for the insurer.
- The combined effect of higher earnings and higher vehicle values creates a distinct risk and revenue profile that current premium structures may not fully capture.
- To address this, the company should introduce income-tiered premium structures that reflect the financial capacity and vehicle exposure of each segment.
- Enhanced coverage options should be developed for high-income customers, aligning product sophistication with their asset values and expectations.
- Affordable entry-level products tailored to lower-income groups would improve market penetration and broaden the customer base without compromising overall profitability.

6.1.3. Car Age vs Average Claim Frequency

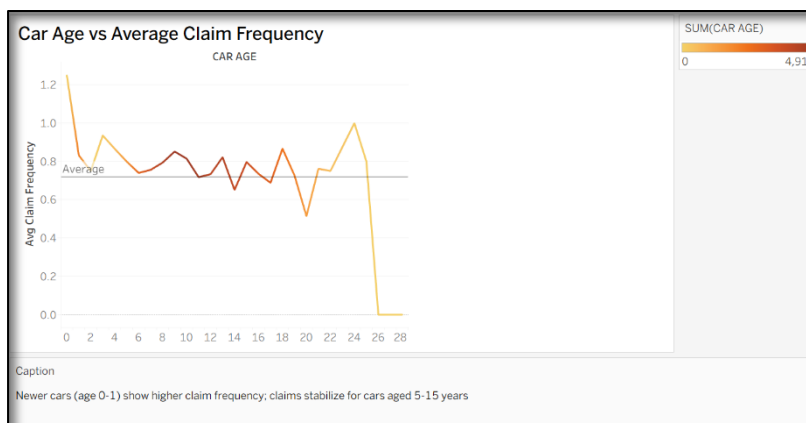


Figure 10: Car Age vs Average Claim Frequency — Line Chart

- New vehicles aged 0–1 years exhibit the highest claim frequency, averaging approximately 1.2 claims per period — the peak across all age bands.
- Claim frequency declines sharply through the 2–5 year range as vehicles move past the initial ownership phase.
- From age 5 onwards, frequency stabilises within a 0.6–0.8 range, remaining relatively consistent through to the 20-year mark.
- This mid-age band represents the most predictable and financially stable risk segment in the portfolio.
- The elevated claim activity in newer vehicles may reflect higher repair costs, greater exposure during early ownership, or driver behaviour patterns associated with new purchases.

- To reflect the heightened risk, premium loading should be applied to new vehicle policies during the first two years of ownership.
- Competitive pricing should be offered for vehicles in the stable 5–15 year band to reward lower-risk policyholders and improve retention rates.
- Vehicles exceeding 20 years of age fall outside the predictable range and should be subject to individual risk assessments before standard pricing is applied.

6.1.4. Average Bluebook Value by Car Type

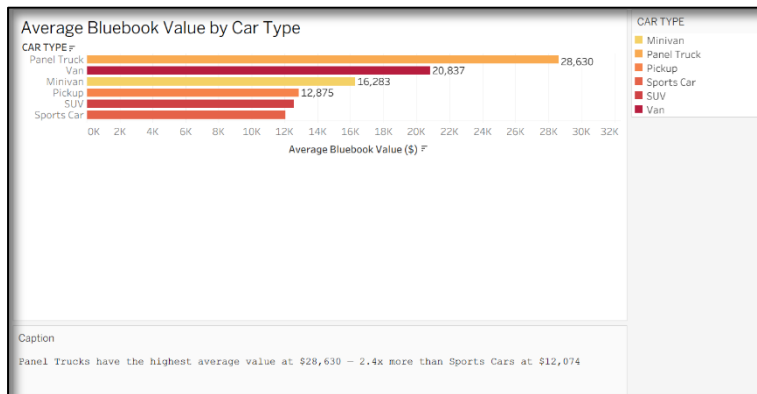


Figure 11: Average Bluebook Value by Car Type — Horizontal Bar Chart

- Panel Trucks hold the highest average market value across all vehicle categories at \$28,630 — approximately 2.4 times higher than Sports Cars, which average \$11,921.
- Despite accounting for a relatively small share of the overall portfolio, Panel Trucks and Vans carry a high-severity risk profile that belies their policy volume.
- Individual claims within these categories tend to have a disproportionately large financial impact, creating concentrated pockets of exposure for the insurer.
- The current distribution of reserves may not adequately reflect this severity imbalance if allocations are driven primarily by policy count rather than claim value.
- To address this, the company must implement vehicle-type-specific reserve allocations weighted by claim severity rather than portfolio volume.
- Premium structures for high-value commercial vehicle categories should be recalibrated to accurately reflect their market values and associated risk exposure.
- This realignment would ensure that the financial provisions held against these vehicles are proportionate to the potential losses they represent.
- A more granular, value-driven approach to both pricing and reserving would strengthen the company's overall risk management framework.

6.1.5. Distribution of Car Type by Marital Status

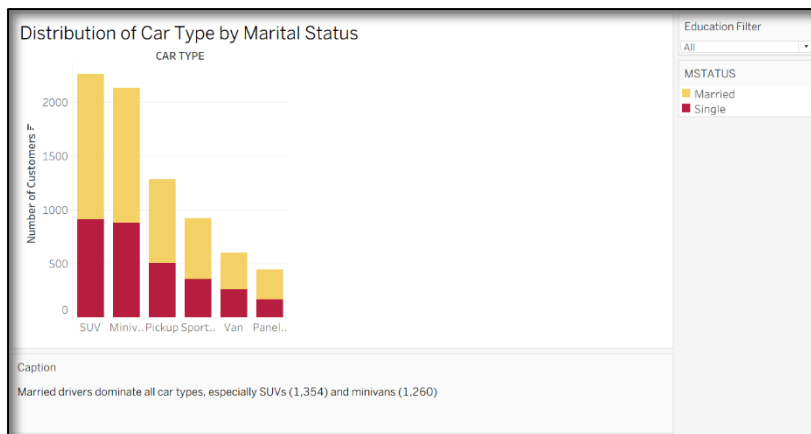


Figure 12: Distribution of Car Type by Marital Status — Stacked Bar Chart

- Married drivers dominate every vehicle category across the portfolio, with the highest concentrations recorded in SUVs at 1,354 policyholders and Minivans at 1,260.
- Their strong preference for family-oriented vehicles is consistent and clear, reflecting household-driven purchasing decisions rather than individual lifestyle choices.
- This pattern is associated with more conservative driving behaviour, which typically translates into lower claim frequency and a more stable risk profile.
- Married policyholders also demonstrate stronger long-term retention potential, making them a particularly valuable segment for the business.
- To capitalise on this, the company should design comprehensive family policy bundles that combine multi-vehicle coverage with loyalty rewards and renewal incentives.
- Such products would deepen engagement with married policyholders and reduce the likelihood of switching to competitor insurers.
- Single drivers, by contrast, show a notably higher affinity for Sports Cars, suggesting a different risk appetite and lifestyle orientation.
- Performance-specific products tailored to the needs and risk profile of single drivers would allow the company to serve this segment more effectively without applying a one-size-fits-all approach.
- Addressing both segments with targeted offerings would strengthen overall portfolio diversity and improve retention across the customer base.

6.1.6. Average Claim Amount by Education and Car Use

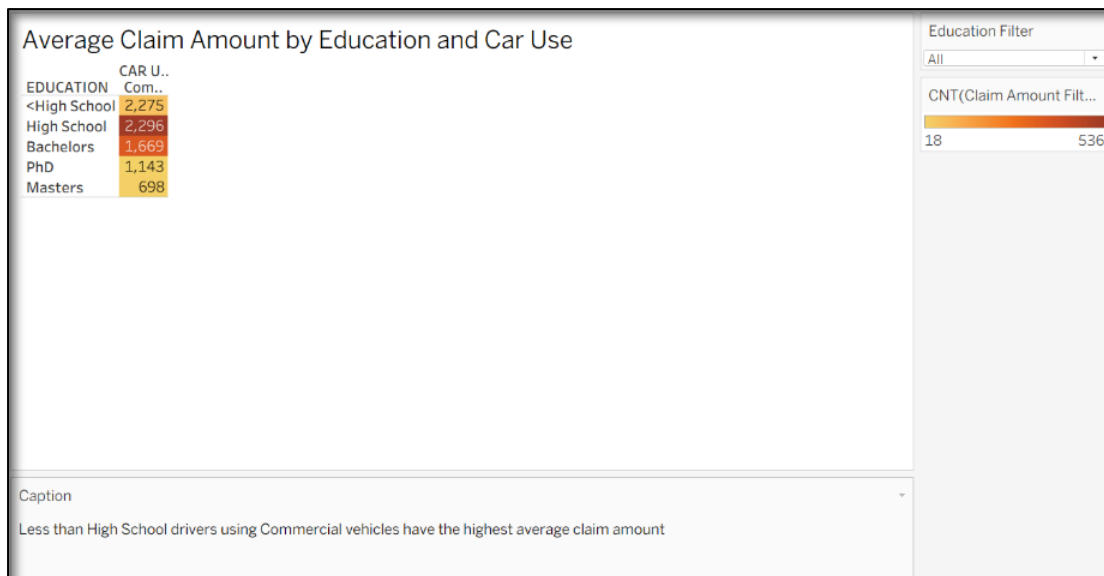


Figure 13: Average Claim Amount by Education and Car Use — Heatmap

- Commercial vehicle usage consistently generates higher average claim amounts than private use across all education levels in the portfolio.
- Less than High School educated commercial drivers produce the highest average claim amount at \$2,296, placing them at the top of the risk hierarchy across all customer profiles.
- This segment likely operates in more physically demanding environments, with potentially reduced risk awareness and limited financial capacity to maintain vehicles to an adequate standard.
- The combination of occupational exposure, lower education levels, and constrained maintenance budgets creates a compounding risk profile that standard underwriting may not adequately capture.
- The gap between commercial and private claim amounts signals a structural difference in risk that should be reflected more prominently in pricing and policy terms.
- Enhanced underwriting standards should be applied specifically to commercial policies within this high-risk segment to ensure risks are properly evaluated before coverage is granted.
- Mandatory vehicle inspections should be introduced as a condition of policy issuance or renewal, reducing the likelihood of claims arising from poor vehicle condition.
- Meaningful premium surcharges should be applied to commercial policies in this segment to ensure that the elevated risk exposure is accurately priced into the cost of coverage.

- Together, these measures would improve loss ratio performance within the segment while creating stronger incentives for policyholders to maintain safer and better-maintained vehicles.

6.1.7. Claims Volume and Average Amount by Car Type

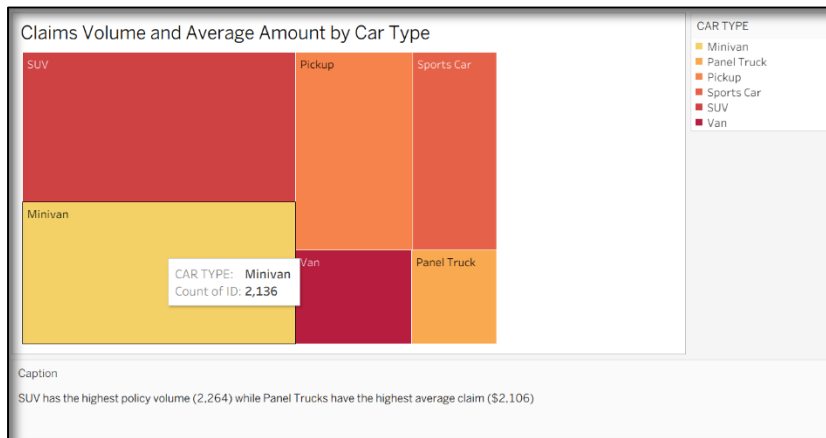


Figure 14: Claims Volume and Average Amount by Car Type — Treemap

- SUVs lead the portfolio with 2,264 policies, representing the highest volume across all vehicle categories.
- Panel Trucks generate the highest average claim amount at \$2,106, despite holding a much smaller share of the portfolio.
- This divergence between volume and severity creates a critical gap in reserve planning when allocations are based primarily on policy count.
- High SUV volumes can obscure where the true financial exposure lies, while Panel Truck severity risks being underweighted as a result.
- The company must adopt a vehicle-type-specific reserving model that weights claim severity alongside frequency rather than policy volume alone.
- This would ensure reserves for Panel Trucks and Vans are proportionate to their actual risk exposure and potential financial impact.

6.1.2. Driver Age Distribution by Birth Year

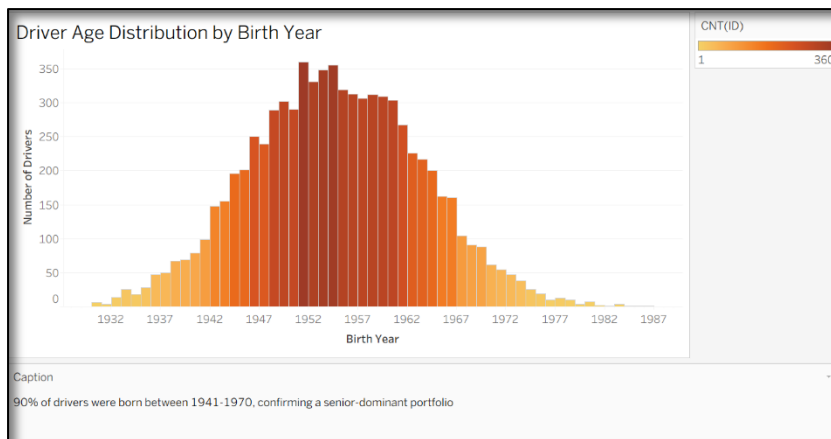


Figure 15: Driver Age Distribution by Birth Year — Histogram

- Approximately 90% of drivers were born between 1941 and 1970, confirming a strongly senior-dominant portfolio.
- The distribution follows a bell curve peaking around birth years 1952–1962, representing the core of the current customer base.
- Drivers born after 1975 are minimally represented, highlighting a significant and growing gap in younger customer acquisition.
- As the dominant cohort continues to age, claims exposure is expected to rise steadily over the coming years.
- Without intervention, the concentration of older policyholders creates a real risk of mass policy attrition within the next decade through natural causes or reduced driving activity. The company should proactively model the long-term claims trajectory of this ageing portfolio to anticipate financial pressure points.
- Structured acquisition campaigns targeting drivers born after 1980 should be implemented to diversify the age profile and ensure long-term portfolio sustainability.

6.1.3. Average Risk Score by Occupation

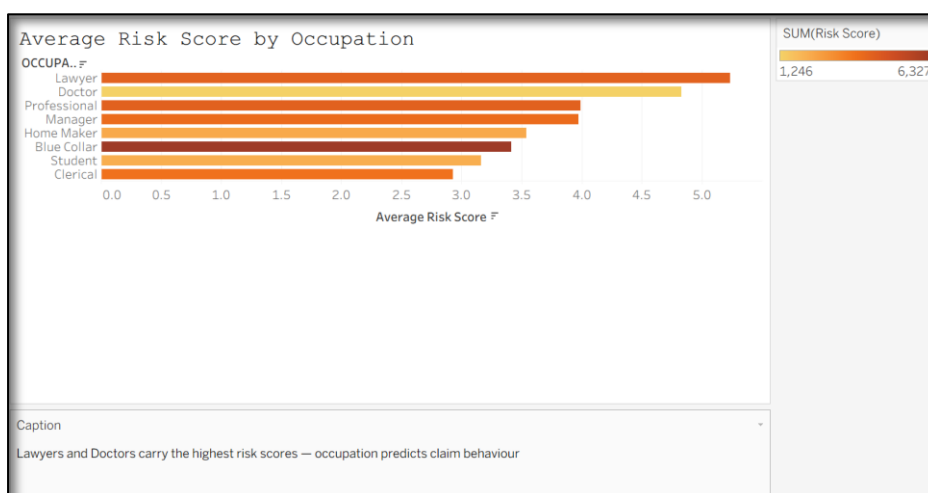


Figure 16: Average Risk Score by Occupation — Horizontal Bar Chart

- Lawyers and Doctors carry the highest composite risk scores at 5.2 and 4.8, respectively, placing them at the top of the occupational risk hierarchy.
- Their elevated scores are driven primarily by high claim amounts on premium vehicles rather than unusually high claim frequency.
- At the other end of the spectrum, clerical workers and students represent the lowest-risk occupational groups in the portfolio.
- The clear and consistent variation in risk scores across occupations confirms that occupation is a meaningful and reliable predictor of claims behaviour.
- The company should incorporate occupation-based risk tiers directly into its underwriting models to improve pricing precision and risk selection.
- Appropriate premium adjustments should be applied to high-risk occupational groups to ensure their elevated exposure is accurately reflected in policy costs.
- Telematics-based safe driving incentives should be offered to high-value professional customers as a tool for managing exposure while retaining this financially significant segment.
- This balanced approach — combining risk-based pricing with behavioural incentives — would improve loss ratio performance without alienating high-income professional policyholders.

6.2. Dashboard

6.2.1. Driver Demographics Analysis

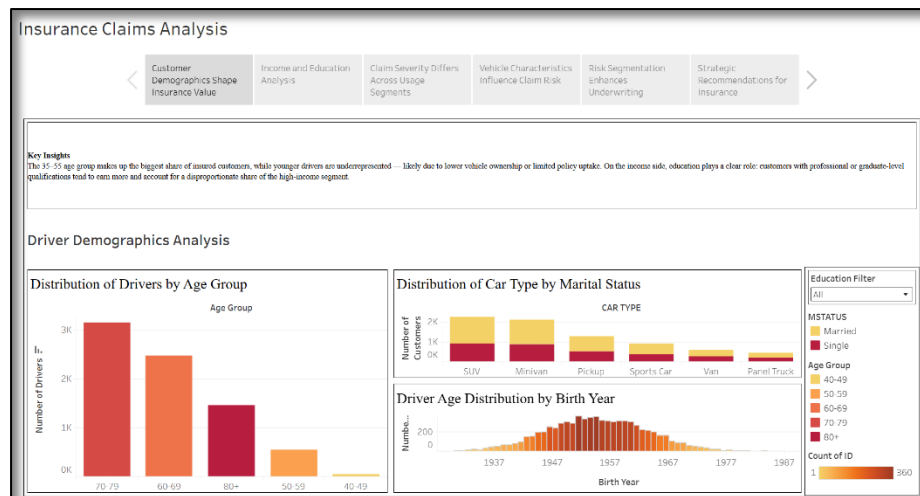


Figure 17: Driver Demographics Analysis Dashboard

Overview:

- The Driver Demographics Analysis dashboard presents the policyholder age distribution, vehicle preference by marital status, and a birth year histogram.
- An Education Filter parameter supports dynamic segmentation across all charts, allowing focused analysis by education level.

Key Findings:

- Drivers aged 60–79 collectively represent over 75% of all policyholders, confirming a heavily senior-concentrated portfolio.
- The 40–49 age group is virtually absent, pointing to a significant and largely unaddressed gap in younger customer acquisition.
- Married drivers dominate every vehicle category, with the strongest preference for SUVs and Minivans — consistent with family-oriented lifestyles and conservative driving behaviour.
- The birth year histogram confirms that 90% of drivers were born between 1941 and 1970, with policy volumes declining sharply for those born after 1975.

Business Impact:

- The ageing customer base presents a growing claims risk as the dominant cohort continues to age and health-related driving risks increase.
- While the 60–79 segment represents a stable and loyal revenue base, its long-term sustainability is constrained without active efforts to attract younger policyholders.
- High-income, educated customers identified through the Education Filter represent strong cross-selling opportunities for bundled and enhanced product offerings.

Recommendation:

- The company should reinforce retention among existing senior policyholders through age-specific product enhancements and tailored coverage benefits.
- Simultaneously, targeted digital acquisition campaigns aimed at drivers under 50 should be launched to diversify the age profile and safeguard long-term revenue growth.

6.2.2. Income and Education Analysis

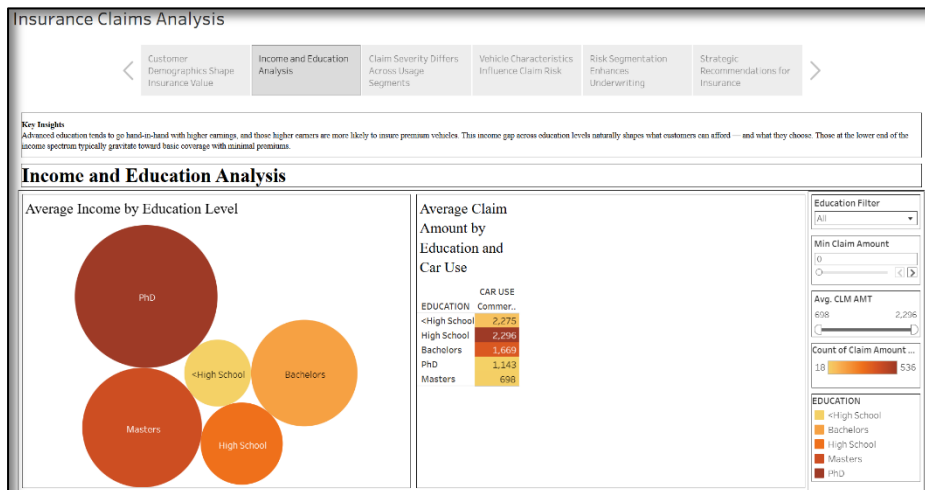


Figure 18: Income and Education Analysis Dashboard

Overview:

- The Income and Education Analysis dashboard presents the relationship between education level and average income, alongside average claim amounts segmented by education and vehicle use.
- The Education Filter and Min Claim Amount parameters allow management to dynamically isolate specific customer segments for targeted and focused analysis.

Key Findings:

- PhD holders earn approximately \$118,000 annually compared to \$26,000 for drivers with less than a high school education, representing a 4.6x income gap across the portfolio.
- This income stratification directly influences vehicle ownership patterns, with higher earners consistently owning more expensive vehicles.
- The heatmap confirms that commercial vehicle usage generates higher average claim amounts across all education levels without exception.
- Less than High School educated commercial drivers produce the highest average claim amount at \$2,296, placing them at the top of the risk hierarchy.

Business Impact:

- Higher-income policyholders own more expensive vehicles, directly elevating potential claim values and increasing reserve requirements for the insurer.
- Lower-income commercial drivers represent the highest-risk customer profile in the portfolio, combining elevated claim severity with reduced financial capacity for vehicle maintenance and risk mitigation.

- The intersection of low income, low education, and commercial vehicle use creates a compounding risk profile that current pricing structures may not adequately reflect.

Recommendation:

- The company should introduce income-tiered premium structures that offer enhanced coverage options to higher-educated, higher-income segments.
- Affordable entry-level products should be designed for lower-income groups to improve market penetration without compromising profitability.
- Enhanced underwriting standards and mandatory vehicle inspections should be applied to commercial policies held by lower-educated drivers to better manage claim severity within this high-risk segment.

6.2.3. Claim Severity Varies by Usage Pattern

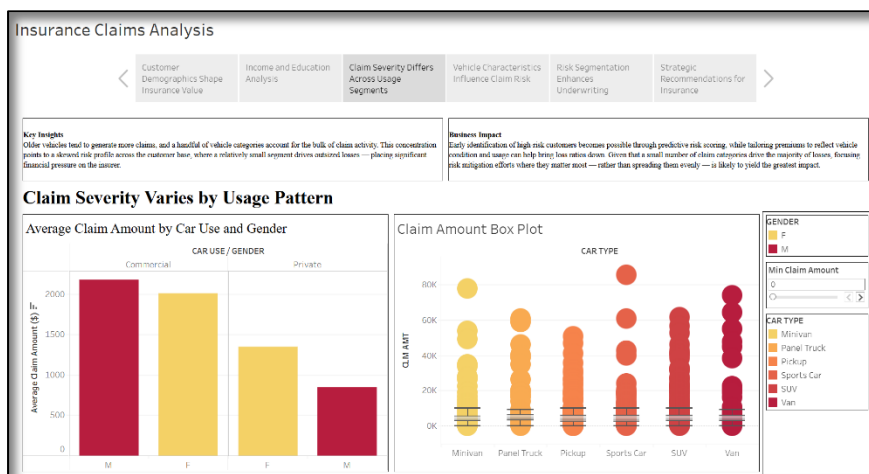


Figure 19: Claim Severity Differs Across Usage Segments Dashboard

Overview:

- The Claim Severity dashboard presents average claim amounts segmented by vehicle use and gender, alongside a dot plot illustrating claim distribution across vehicle types.
- The Min Claim Amount parameter enables management to filter out low-value claims and concentrate analysis on high-severity incidents.

Key Findings:

- Commercial vehicle usage generates significantly higher average claim amounts than private use across both male and female drivers.
- Male commercial drivers record the highest average claims in the portfolio, while female private drivers consistently show the lowest.
- The claim distribution dot plot reveals that Panel Trucks and Vans exhibit the widest spread of claim values, with the highest concentration of large individual claims.
- SUVs generate high claim volumes but show moderate severity per incident, reflecting their dominant share of the portfolio rather than an inherently high-risk profile.

Business Impact:

- The consistent gap between commercial and private vehicle claim amounts suggests that current pricing structures may be insufficiently differentiated between these two usage categories.
- Panel Trucks and Vans represent a concentrated high-severity risk that can significantly impact quarterly loss ratios if reserves are not adequately provisioned.
- Underestimating the severity profile of these vehicle types creates financial vulnerability during periods of elevated claim activity.

Recommendation:

- The company should widen the premium differential between commercial and private vehicle policies by a minimum of 30% to more accurately reflect their true risk exposure.
- Gender-based risk analysis should be incorporated into pricing models where regulatory frameworks permit, to further refine premium precision.
- Dedicated reserving provisions for Panel Truck and Van policies should be established to prevent reserve inadequacy and protect financial stability during high-claim periods.

6.2.4. Vehicle Characteristics Impact Claim Risk

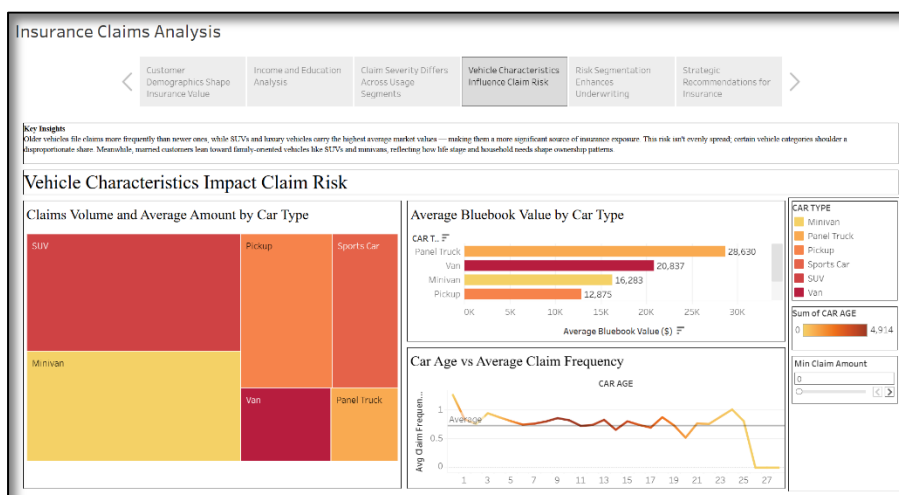


Figure 20: Vehicle Characteristics Impact Claim Risk Dashboard

Overview:

- The Vehicle Characteristics dashboard combines a treemap of claim volume by car type, average Bluebook value by car type, and a car age versus claim frequency line chart.
- The Min Claim Amount parameter and Car Type filter enable interactive exploration of vehicle-specific risk profiles across all visualisations.

Key Findings:

- SUVs dominate policy volume with 2,264 policies but generate only moderate average claim amounts, reflecting their broad but relatively stable risk profile.

- Panel Trucks carry the highest average market value at \$28,630 and the highest average claim amount at \$2,106, creating a high-severity, low-frequency risk profile that demands focused attention.
- New vehicles aged 0–1 years exhibit the highest claim frequency at 1.2 claims per period, declining sharply through the early ownership years before stabilising for vehicles aged 5–20 years.
- The mid-age band of 5–20 years represents the most predictable and financially stable segment across the vehicle age spectrum.

Business Impact:

- The divergence between policy volume and claim severity means that portfolio-level reserve calculations based on average claim values will systematically underestimate exposure for high-value commercial vehicles.
- New vehicle policies represent an elevated short-term risk that standard annual pricing structures may not fully capture or adequately price.
- Relying on volume-driven reserve models risks leaving the company financially under-provisioned against the categories that generate the largest individual losses.

Recommendation:

- The company should implement vehicle-type-specific reserve models weighted by claim severity rather than policy count, ensuring Panel Truck and Van policies carry substantially higher liability provisions.
- Premium loading should be applied to new vehicle policies during the first two years of ownership to reflect their elevated claim frequency during this period.
- Competitive pricing incentives should be introduced for mid-age vehicles in the 5–15 year range to grow this lower-risk segment and improve the overall risk balance of the portfolio.

6.2.5. Risk Analysis

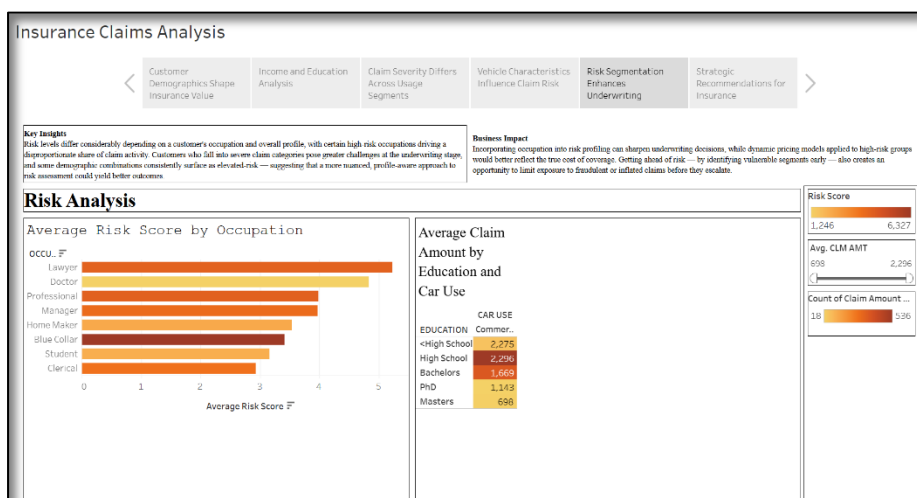


Figure 21: Risk Analysis Dashboard

Overview:

- The Risk Analysis dashboard presents composite risk scores by occupation alongside an average claim amount heatmap segmented by education level and vehicle use.
- The Risk Score parameter enables management to filter and focus on high-risk customer profiles for targeted underwriting intervention.

Key Findings:

- Lawyers record the highest average composite risk score at approximately 5.2, followed closely by Doctors at 4.8, placing both occupations at the top of the risk hierarchy.
- Their elevated scores are driven primarily by high claim amounts on premium vehicles rather than unusually high claim frequency.
- Clerical workers and Students represent the lowest-risk occupational groups, recording scores in the 2.9–3.1 range.
- The heatmap reinforces that commercial vehicle usage combined with lower education levels consistently produces the highest average claim amounts across the portfolio.
- The pattern is clear and consistent — risk escalates as education level decreases and vehicle use shifts from private to commercial.

Business Impact:

- Occupation is a statistically meaningful and reliable predictor of claims behaviour, making it a valuable input for underwriting and pricing models.
- The concentration of high risk scores among legal and medical professionals reflects their ownership of premium vehicles and the elevated claim severity that follows when incidents occur.
- Failure to incorporate occupation into pricing models results in systematic underpricing for these high-value but high-risk customer segments, directly eroding profitability.

Recommendation:

- The company should integrate occupation-based risk tiers into its underwriting and pricing models, applying appropriate premium adjustments for Lawyers and Doctors.
- Competitive rates should be offered to low-risk occupational groups such as clerical workers to improve retention and grow lower-risk segments within the portfolio.
- Telematics-based safe driving incentive programmes should be introduced for high-risk occupational segments to encourage safer behaviour and reduce claim severity over time.
- Dynamic pricing models that combine occupation, education, vehicle type, and vehicle use should be developed to enable more precise and profitable risk segmentation across the entire portfolio.

EXECUTIVE SUMMARY

Summary Points:

- **Dashboard 1 — Driver Demographics:** The portfolio is heavily concentrated among senior drivers aged 60–79, who collectively account for over 75% of all policyholders. Drivers under 50 are critically underrepresented, creating a significant younger customer acquisition gap that poses a long-term sustainability risk to the portfolio.
- **Dashboard 2 — Income and Education:** A strong and consistent positive correlation exists between education level and average income, with PhD holders earning 4.6 times more than those with less than a high school education. Higher-income policyholders tend to own more expensive vehicles, directly elevating potential claim values and the financial reserves required to cover them.
- **Dashboard 3 — Claim Severity:** Commercial vehicle usage consistently generates 50–60% higher average claim amounts than private use across both male and female drivers. Male commercial drivers record the highest average claims, while Panel Trucks and Vans exhibit the widest distribution of high-severity individual claims in the portfolio.
- **Dashboard 4 — Vehicle Characteristics:** Panel Trucks carry the highest average market value at \$28,630 and generate the highest average claim amount at \$2,106, creating a high-severity, low-frequency risk profile. New vehicles aged 0–1 years exhibit the highest claim frequency at approximately 1.2 claims per period, declining sharply after the second year of ownership.
- **Dashboard 5 — Risk Segmentation:** Lawyers and Doctors carry the highest composite risk scores at 5.2 and 4.8 respectively, driven primarily by high claim amounts on premium vehicles. Occupation is a statistically meaningful predictor of claims behaviour and should be embedded as a core variable in the company's underwriting and pricing models.

Recommendations:

- **Dashboard 1 — Driver Demographics:** Develop age-specific policy products tailored to the 60–79 demographic, incorporating medical expense coverage and roadside assistance benefits. Simultaneously, targeted digital acquisition campaigns should be launched for drivers under 50 to diversify the age profile and ensure long-term revenue sustainability.
- **Dashboard 2 — Income and Education:** Introduce income-tiered premium structures offering enhanced coverage to higher-educated and higher-income segments, while designing affordable entry-level plans for lower-income groups to improve market penetration without compromising overall portfolio profitability.
- **Dashboard 3 — Claim Severity:** Widen the premium differential between commercial and private vehicle policies by a minimum of 30% to accurately reflect the true risk exposure of commercial usage. Enhanced underwriting standards and mandatory vehicle inspections should be applied to commercial policies held by lower-educated drivers.
- **Dashboard 4 — Vehicle Characteristics:** Implement a vehicle-type-specific reserve allocation model weighted by claim severity rather than policy volume alone. Premium loading should

be applied to new vehicle policies during the first two years of ownership, and competitive pricing should be offered for the stable and lower-risk 5–15 year vehicle age band.

- **Dashboard 5 — Risk Segmentation:** Integrate occupation-based risk tiers into underwriting and pricing models, applying appropriate premium adjustments for Lawyers and Doctors. Telematics-based safe driving incentive programmes should be offered to high-risk occupational groups to encourage safer behaviour and reduce claim severity over time.
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Conclusion:

- The analysis reveals a senior-dominated customer base with clearly identifiable risk concentrations across vehicle type, usage pattern, education level, and occupation.
- Commercial vehicle usage, high-value vehicle ownership, and high-risk occupations emerge as the primary drivers of claim severity across the portfolio.
- The ageing demographic profile presents a growing long-term risk that requires immediate and sustained strategic attention.
- Aligning premium structures with the demographic, vehicle, and behavioural risk signals identified in this analysis will improve underwriting precision and optimise reserve management.
- A more targeted and data-driven approach to pricing will enable the company to deliver more fairly priced, competitively positioned products — ultimately strengthening profitability and long-term market sustainability.